

# Reference-Set Decoupled Co-Adaptive Training for Physics-Informed Neural Networks

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## Abstract

Adaptive collocation changes the training measure observed by global loss controllers in physics-informed neural networks (PINNs). We introduce and evaluate reference-set decoupled co-adaptive training (ReCoA-PINN), which estimates global component weights on a fixed reference set while optimizing the network on adaptive collocation points. A controlled study compares this mechanism with coupled control, proposal correction, kernel-density volume weighting and variational residual attention on a common PyTorch backbone. Historical paired experiments do not establish general stabilization or accuracy gains from the fixed reference. We extend the evaluation with 200 V09 ablations and 560 V10 runs covering difficult observations, doubled optimization budgets and common-loss comparisons. Independent Cole–Hopf validation identifies a maximum discrepancy of 0.28578 in the historical coarse Burgers reference. A Huber-versus-MSE confound is addressed through new MSE-matched experiments. In the corrected comparison, variational residual attention has the lowest median relative L2 error in four observation regimes; 11 of 16 paired L2 contrasts pass Holm correction, while no maximum-error contrast passes its separate correction. These exploratory findings support configuration-dependent conclusions. We distinguish the finite-population identity of fresh importance-weighted samples from claims about the complete adaptive training trajectory, and document the controls needed to assess reference-set decoupling without assuming its superiority.

**Keywords:** physics-informed neural networks; adaptive collocation; loss balancing; reference sets; importance weighting; reproducibility.

## 1 Introduction

Physics-informed neural networks (PINNs) approximate a physical field while penalizing the residual of its governing differential equation. Their training objective generally combines the physics residual, initial and boundary conditions, and, when available, observations. Because these components can have different scales and convergence rates, adaptive global loss balancing is frequently used. Residual-based adaptive sampling is also used to allocate more collocation points to poorly resolved regions.

Each mechanism is reasonable in isolation, but their joint use raises a subtle question. If the global

loss controller estimates the importance of the physics term on the same nonuniform points selected by the residual sampler, it no longer observes the original uniform-domain objective. The controller and the sampler may therefore reinforce one another. This interaction is distinct from the well-known imbalance between loss components: it concerns the measure under which a component is evaluated.

Previous work has already established adaptive loss balancing, residual-based sampling, importance sampling, volume correction and variational residual attention. Therefore, our aim is not to claim the invention of these individual mechanisms. We conduct a controlled study of competing training measures: retaining the coupled controller, estimating global weights on a fixed reference set, correcting a known discrete proposal, estimating local volume by KDE, or deliberately tilting the objective through vRBA.

The study asks five questions:

1. Does residual sampling measurably alter the component losses and gradients observed by a global controller?
2. Does a fixed, independent reference set improve training stability or solution accuracy?
3. Does exact proposal correction improve a matched uncorrected adaptive sampler under sparse and noisy data?
4. Are the conclusions robust to PDE family and to an established alternative, KDE volume weighting?
5. Does a deliberately tilted variational objective improve global and local errors consistently on the same backbone?

Our principal finding is conditional. The feedback is measurable and exact correction can be beneficial. vRBA performs favorably in historical relative-L2 comparisons, but a common-loss audit qualifies the noisy-data claim; maximum error and cost do not show universal dominance. This mixture of positive, null and negative results identifies the controls required before an adaptive PINN method can credibly claim robustness.

## 2 Related work and positioning

Raissi, Perdikaris and Karniadakis established the modern PINN formulation for forward and inverse differential-equation problems (Raissi, Perdikaris, and Karniadakis 2019). Gradient-flow pathologies and adaptive balancing were subsequently analyzed by Wang, Teng and Perdikaris and by several self-adaptive weighting methods (Wang, Teng, and Perdikaris 2021). Wu et al. provided a broad comparison of nonadaptive and residual-based adaptive collocation (Wu et al. 2023). Nabian, Gladstone and Meidani introduced importance sampling for more efficient PINN training (Nabian, Gladstone, and Meidani 2021).

The closest antecedent to the measure question is VW-PINNs by Song et al. They show that nonuniform collocation can make the standard empirical PDE loss ill-conditioned and propose weighting residuals by local volumes estimated via kernel density estimation (Song et al. 2025). Recent work has gone further: joint weighting and sampling methods explicitly combine the two mechanisms, while the vRBA framework of Toscano et al. interprets residual adaptivity through convex potentials and variational objectives (Toscano et al. 2026).

These studies prevent two overly broad claims: combining loss weighting and adaptive sampling is not new, and correcting the effect of a nonuniform collocation measure is not new. The potentially useful contribution here is a paired controlled design that contrasts a fixed controller reference, a known-proposal correction, KDE volume correction, a variational tilted objective and matched controls under exactly the same PyTorch backbone.

Chen, Howard and Stinis combine residual-decay weighting with adaptive sampling (Chen, Howard, and Stinis 2025). Singh et al. combine smoothed gradient balancing and residual-based collocation on Burgers and Allen–Cahn (Singh et al. 2026). These are direct antecedents to a broad co-adaptation claim. Our M5 comparator is not a faithful reproduction of either full published protocol. The present contribution is the explicit test of a fixed controller reference together with matched mechanism controls and documented limitations.

Modern point-placement alternatives include PINNACLE, which selects collocation and experimental points jointly (Lau et al. 2024), QR-DEIM-based collocation selection (Celaya, Fuentes, and Riviere 2025), and gradient-based point movement in PACMANN (Visser, Heinlein, and Giovanardi 2025). They are discussed but not evaluated here. Consequently this study does not establish state-of-the-art superiority over contemporary placement methods.

### 3 Methods

#### 3.1 Base objective

For model parameters  $\theta$ , the total objective is

$$\mathcal{L}(\theta) = \sum_{j \in \{f, i, b, d\}} \lambda_j \mathcal{L}_j(\theta), \quad (1)$$

where  $f$ ,  $i$ ,  $b$ , and  $d$  denote the physics, initial-condition, boundary-condition and observation terms. The standard physics term on collocation points  $z_i$  is

$$\mathcal{L}_f = \frac{1}{N} \sum_{i=1}^N r_\theta(z_i)^2. \quad (2)$$

#### 3.2 Controlled variants

M0 uses uniform collocation and fixed component weights. M1 adds adaptive component weights. M3 adds residual sampling but retains fixed component weights. M5 combines both mechanisms and computes the controller update on the adaptive training set. ReCoA-PINN (M6) uses the same adaptive-sampling protocol as M5 but computes the controller update on a fixed independent audit set of equal cardinality.

M7U samples with replacement from a finite candidate pool according to the residual proposal  $q$ , but leaves the physics loss uncorrected. M7 uses the same sampling mechanism and random-number streams, and attaches the Hansen–Hurwitz weight

$$w_i = \frac{1}{N_c q_i}, \quad \widehat{\mathcal{L}}_{f, \text{M7}} = \frac{1}{N} \sum_i w_i r_\theta(z_i)^2, \quad (3)$$

where  $N_c$  is the candidate-pool size. Retained points keep the proposal weight assigned when they were sampled. A uniform proposal floor bounds the largest possible importance weight.

For VW and VW-CA, a Gaussian KDE estimates the density of the realized collocation set after mapping all coordinates to the unit hypercube. With  $V_i \propto 1/\hat{p}_h(z_i)$ , the physics loss is

$$\mathcal{L}_{f, \text{VW}} = \frac{\sum_i V_i^2 r_\theta(z_i)^2}{\sum_i V_i^2}. \quad (4)$$

VW uses fixed global component weights. VW-CA uses the same adaptive global controller as M5 and M7. The default KDE bandwidth follows Scott’s rule; separate runs use 0.5, 1 and 2 times this bandwidth.

For vRBA, fixed point sets receive persistent local multipliers. With the exponential potential, the numerically stable normalized attention target is

$$\hat{q}_i^k = \exp\left(\frac{|r_i| - \max_j |r_j|}{\epsilon_k}\right), \quad \epsilon_k = \frac{c \max_j |r_j|}{\log(k+2)}. \quad (5)$$

The local update and component loss are

$$\lambda_i^{k+1} = \gamma_k \lambda_i^k + \eta (\phi \hat{q}_i^k + 1 - \phi), \quad \mathcal{L}_j^{\text{vRBA}} = \frac{1}{N_j} \sum_i (\lambda_{j,i} r_{j,i})^2. \quad (6)$$

The physics component keeps global weight one; other component weights track ratios of exponentially smoothed gradient norms. We use the published first-order defaults (`eta=0.01`, `phi=0.8`, local cap schedule, gradient EMA `0.99`, global-weight EMA `0.99975`). This is a controlled PyTorch common-backbone implementation of the vRBA mechanism, not an exact reproduction of the authors’ JAX architecture, hard constraints, optimizer sequence or training horizon.

### 3.2.1 Fixed-reference controller and interpretation

Let  $R$  be the fixed reference set and  $S_k$  the adaptive collocation set. ReCoA-PINN computes controller losses and gradient norms on  $R$  while updating the network using  $S_k$ . M5 uses  $S_k$  for both. The physics and known-condition reference points are drawn independently at initialization; observed sensors and their targets are shared. The reference set is used repeatedly to control training and is therefore not an independent test set for the trained model. Adaptive trajectories can subsequently diverge even with identical random seeds.

For component  $j$ , controller updates implement

$$g_{j,k} = \|\nabla_\theta L_j\|_2, \quad \bar{g}_{j,k} = \beta \bar{g}_{j,k-1} + (1 - \beta) g_{j,k}, \quad a_{j,k} = \frac{\text{mean}_\ell \bar{g}_{\ell,k}}{\bar{g}_{j,k}} \left( \frac{L_{j,k}}{L_{j,0}} \right)^\alpha. \quad (7)$$

The implementation normalizes  $a_{j,k}$  to unit mean, clips it between  $w_{\min}$  and  $w_{\max}$ , renormalizes the clipped targets to unit mean and then applies an EMA to the component weights. Here  $\beta = 0.95$ ,  $\alpha = 0.5$ ,  $w_{\min} = 0.1$  and  $w_{\max} = 10$ . Clipping occurs before the second normalization: these settings are not strict final-weight bounds. Positive numerical floors prevent division by zero. This paper evaluates the implemented rule; it does not claim a convergence theorem for the controller.

At each iteration, the algorithm (i) refreshes the scheduled fraction of collocation points if due, (ii) updates global component weights from the method-specific reference if due, (iii) constructs the weighted training loss, and (iv) takes one Adam step with gradient clipping. Evaluation points are never used for these updates. For vRBA, local attention and its global controller are updated on fixed point sets; its ablations are not ablations of adaptive point placement or of the M6 reference mechanism.

### 3.2.2 Scope of exact proposal correction

For a frozen model  $\theta$  and candidate population  $C$ , a fresh with-replacement draw  $I \sim q$  satisfies

$$\mathbb{E}_{I \sim q} \left[ \frac{r_{\theta}(z_I)^2}{N_c q_I} \mid \theta, C, q \right] = \frac{1}{N_c} \sum_{i=1}^{N_c} r_{\theta}(z_i)^2. \quad (8)$$

This is a finite-population quadrature identity, not a guarantee for the full adaptive optimization path. Retained points carry weights from earlier candidate populations, and later model parameters depend on earlier samples. The identity does not prove conditional unbiasedness of every later gradient or elimination of continuous-domain discretization error. M7 and M7U share the sampling algorithm and seeded random streams, but their model-dependent proposals and realized point sets need not remain identical after training diverges. The corrected-versus-uncorrected contrast concerns the full paired training trajectories.

## 3.3 Problems

The benchmark suite contains viscous Burgers, a manufactured Allen–Cahn problem, a standing-wave equation and a manufactured two-dimensional Helmholtz problem. An initial high-frequency Helmholtz screen was not used for method ranking because every method had relative error above one. We calibrated a solvable configuration using only M0 and a predetermined criterion  $L_2 < 0.2$ , then froze it before comparing adaptive methods.

Four Burgers observation regimes assess robustness: thirteen clean observations; thirteen observations with Gaussian noise and outliers; one observation with strong heteroscedastic noise; and a missing sensor block with correlated noise.

The governing equations and exact or independent targets are:

- **Burgers:**  $u_t + uu_x - \nu u_{xx} = 0$ , with  $\nu = 0.01/\pi$ ,  $x \in [-1, 1]$ ,  $t \in [0, 1]$ ,  $u(0, x) = -\sin(\pi x)$  and zero Dirichlet boundaries. V10 uses the Cole–Hopf integral solution (Burkardt n.d.).
- **Manufactured Allen–Cahn:**  $u_t - Du_{xx} - \rho(u - u^3) = f$ , with  $D = 10^{-3}$ ,  $\rho = 5$ , the same space-time domain and  $u = e^{-t} \sin(\pi x)$ . The forcing is  $f = -u + D\pi^2 u - \rho(u - u^3)$ ; exact initial and boundary values are imposed softly. This is not the difficult unforced benchmark.

- **Wave:**  $u_{tt} - c^2 u_{xx} = 0$ , with  $c = 1$  on  $[0, 1]^2$ ,  $u = \sin(\pi x) \cos(\pi t)$ , zero boundary values, initial displacement  $\sin(\pi x)$  and zero initial velocity. Displacement and velocity MSEs are summed in the initial-condition component.
- **Manufactured Helmholtz:**  $\Delta u + k^2 u = f$  on  $[0, 1]^2$ , with  $k = 1$ ,  $u = \sin(\pi x) \sin(\pi y)$  and  $f = (1 - 2\pi^2)u$ , with zero boundaries.

All final V09–V10 evaluations concern these four PDE families. The damped oscillator mentioned in the original research blueprint is not included in the reported experimental suite. All physical parameters above are prescribed; the present experiments do not identify unknown PDE coefficients from data.

### 3.4 Experimental controls

The historical ten-seed comparisons and the exploratory V09–V10 campaigns use ten paired seeds 11, 22, 33, 44, 55, 66, 77, 88, 99, 111. Methods share architecture, initialization seed, observation realization, collocation cardinality and clean evaluation target. All configurations and environment metadata are stored with each run. Failed runs are retained and reported rather than silently discarded.

The primary metric is relative  $L_2$  error. Secondary metrics include maximum absolute error, component-weight total variation, the total-variation distance between component-gradient distributions on training and audit points, the logarithmic physics-gradient gap, wall time, residual evaluations, importance-weight effective sample size, and KDE pairwise-distance counts.

We report medians, quartiles, paired median differences, 95% bootstrap intervals and two-sided sign tests. A method is not called superior when only its point estimate improves or when the paired interval crosses zero.

### 3.5 V09–V10 extensions and common-loss audit

V09 consists of 200 runs: four PDE configurations, five local/global/potential variants and ten seeds. V10 adds 200 Burgers observation-regime runs, 200 doubled-step runs and 160 common-loss baseline runs. The latter reuse 40 existing V10 vRBA checkpoints for comparison. Each suite freezes its manifest before training but follows earlier empirical findings; these suites are exploratory, not independent confirmations of hypotheses selected using V07 or V09.

The four factorial arms are neither, `global_only`, `local_only` and `full_exp`; `full_quad` changes only the potential relative to `full_exp`. V10 uses MSE for observations in every arm. In historical noisy-data comparisons, local vRBA multipliers triggered a squared residual loss while several non-vRBA methods used Huber. Consequently, those comparisons contrast complete recipes and do not identify an effect of attention alone. V10 synthetic Burgers observations and evaluation targets use the independent Cole–Hopf solution.

For each suite, four predeclared contrasts per case form a family of 16 tests for relative L2, with a separate family for maximum absolute error. We use two-sided exact sign tests, Holm correction, and 10000-resample percentile bootstrap intervals for the paired median difference. Intervals are marginal, not simultaneous. Initialization and observation tensors are fingerprinted; the 400 factorial/budget V10 checkpoints are re-evaluated independently, including a denser grid; the 160 additional baseline checkpoints are independently checked on the original scoring grid.

### 3.6 Architecture, budgets and observation counts

All networks use tanh activations, Adam with learning rate  $10^{-3}$ , float64 arithmetic, gradient clipping at 100 and one numerical CPU thread per process. Table 1 summarizes the four base configurations. Architectures and point cardinalities are matched between methods within each configuration, not globally across all PDEs.

Table 1: Base architectures and optimization budgets.

| Configuration | Hidden layers x width | Physics / initial / boundary points | V09 steps | V10 doubled steps |
|---------------|-----------------------|-------------------------------------|-----------|-------------------|
| Burgers long  | 4 x 48                | 512 / 128 / 128                     | 2500      | 5000              |
| Allen–Cahn    | 3 x 32                | 256 / 64 / 64                       | 600       | 1200              |
| Helmholtz     | 4 x 48                | 512 / 0 / 128                       | 1500      | 3000              |
| Wave          | 3 x 32                | 256 / 64 / 64                       | 600       | 1200              |

The four difficult-observation Burgers cases use 3 x 32 networks, 256 physics, 64 initial and 64 boundary points, and 600 steps. Their realized observation counts are 13 (sparse clean), 13 (sparse noisy), 1 (extreme) and 64 (missing block). The common-loss extension uses MSE throughout. Detailed noise settings and all paired contrasts are supplied in the supplement.

## 4 Results

Sections 4.1–4.6 report historical configurations and references. New V09–V10 evidence follows in Sections 4.7–4.11. Historical noisy-data rankings are recipe comparisons subject to the loss confound described in Section 3.5.

### 4.1 Fixed-reference decoupling

On long Burgers runs, M5 shows a positive concentration–gradient-gap correlation in all ten seeds. Nevertheless, M6 does not show a conclusive improvement in accuracy under the reported paired interval: the paired median difference M6–M5 is  $-0.0291$ , with 95% interval  $[-0.1208, 0.1678]$ . Nor does it consistently reduce weight variation. Because controller cardinality and residual-evaluation counts are matched, this null result cannot be attributed to additional reference points.

### 4.2 Exact correction and its matched control

M7 substantially reduces the mean physics-gradient log-gap relative to M7U on Burgers: paired median difference  $-0.4351$ , 95% interval  $[-0.6182, -0.2520]$ , favorable in all ten seeds. Across the four difficult observation regimes, M7 also improves relative  $L_2$  error over M7U by  $-0.0483$ ,  $-0.0482$ ,  $-0.0229$ , and  $-0.0360$ , respectively; every bootstrap interval excludes zero.

This effect is not universal. On Allen–Cahn, M7 is worse than M7U by a paired median of  $+0.0097$ , with a 95% interval just above zero. Thus, restoring the chosen target measure may improve diagnostic alignment while worsening the optimization trajectory for a particular PDE.

### 4.3 Comparison with KDE volume weighting

On the standing-wave problem, fixed-weight VW is better than M7 in all ten seeds: M7–VW is +0.0280, 95% interval [0.0199, 0.0483]. On calibrated Helmholtz, the ranking reverses: M7–VW is –0.0181, 95% interval [–0.0235, –0.0112], favorable to M7 in all ten seeds. Burgers long and Allen–Cahn show no conclusive difference between M7 and VW.

At the default KDE bandwidth, M7 outperforms VW-CA in each of the four difficult Burgers observation regimes, with paired intervals excluding zero. However, M7 does not outperform the simpler fixed-weight VW in any of those four regimes. The addition of a global adaptive controller therefore does not automatically improve a locally corrected loss.

### 4.4 Bandwidth sensitivity

The KDE choice changes the conclusion. On noisy Burgers, reducing the bandwidth to 0.5 times Scott’s rule yields median errors 0.4742 for VW and 0.5527 for M7. The paired M7–VW difference is +0.0710, 95% interval [0.0161, 0.1165]. At that bandwidth, the previously favorable M7–VW-CA contrast also becomes inconclusive. On the wave equation, VW remains better than M7 at all three tested bandwidth scales.

### 4.5 Computational cost

In a dedicated sequential CPU profile, M7 adds a median 10.5% wall-time cost relative to M5, 7.2% relative to M6 and 4.9% relative to M7U, while retaining the same number of residual evaluations. For 256 collocation points and 600 steps, the dense VW correction additionally evaluates 1,638,400 pairwise KDE distances per run. The correction part of M7 is linear after the proposal is known, whereas dense KDE is quadratic in the number of collocation points. This theoretical advantage has not yet been verified by a dedicated multi-size CPU/GPU profile.

### 4.6 Variational residual-based attention

The protocol-frozen common-backbone vRBA baseline completes all runs without non-finite values. Its final attention effective-sample fraction ranges from 0.785 to 0.990, confirming non-uniform but non-collapsed local weighting.

vRBA obtains median relative L2 errors of 0.2507 on Burgers, 0.0401 on Allen–Cahn, 0.0388 on calibrated Helmholtz, and 0.1745 on the wave equation. Against M7, paired median differences are respectively –0.1770 [–0.2339, –0.0114], –0.0648 [–0.0903, –0.0509], –0.0611 [–0.0690, –0.0496], and –0.1322 [–0.2163, –0.1190]. Against fixed-weight VW, they are –0.1236 [–0.2566, –0.0283], –0.0617 [–0.1269, –0.0267], –0.0844 [–0.0891, –0.0608], and –0.1061 [–0.1502, –0.0836].

The same L2 signal persists across clean sparse data, noisy and outlier- contaminated data, one strongly corrupted observation, and a missing sensor block. However, the extreme one-observation regime has only eight favorable seeds out of ten against M7 and VW, yielding a two-sided sign-test p-value of 0.109 despite bootstrap intervals below zero.

The maximum-error result contradicts a universal robustness interpretation. On long Burgers runs, vRBA’s median maximum error is 1.7209, compared with 1.3429 for M5 and 1.2993 for M6. The paired



vRBA-M5 difference is +0.3336 [0.0265, 0.5694], and vRBA-M6 is +0.4011 [0.2067, 0.5759]. Thus, vRBA improves the global L2 norm while worsening the largest localized error. Observed runtime overhead relative to M5 ranges from 10.4% to 81.8% across PDEs.

Spatial analysis resolves this apparent contradiction. The median across seeds of the gridwise 95th absolute-error percentile is 0.1027 for vRBA, versus 0.6698 for M5 and 0.6907 for M6; the corresponding 99th percentiles are 0.8588, 1.1470 and 1.0516. vRBA therefore improves almost the entire domain while leaving a very narrow late-time error spike near the shock. In the median error field, this spike peaks around  $t = 0.975$ ,  $x = -0.025$ .

## 4.7 Local/global decomposition at the V09 budgets

The 200-run V09 campaign found that full exponential vRBA improves relative L2 against both local-only and global-only variants for the calibrated Helmholtz case ( $k=1$ ). Both contrasts and the additive-scale interaction pass Holm at  $p=0.03125$ . For the wave case, the quadratic potential improves L2 against the exponential potential (Holm  $p=0.03125$ ). No planned Burgers contrast passes the multiplicity correction. These are configuration-specific findings, not evidence of a universally optimal potential or controller.

V09 separately measured 60 CPU windows using three seeds, five variants and four PDEs. Fifty-nine profiling trials completed; the last trial ended after the prespecified timing window. The full exponential variant has median paired per-step cost ratios of about 1.71–1.86 relative to neither. These are local per-step measurements, not time-to-accuracy or equal-time rankings.

## 4.8 Independent Burgers reference

Cole–Hopf quadrature at orders 128 and 256 agrees within  $9.11e-15$  on the tested 161-by-321 grid. Adaptive quadrature independently verifies 20 selected points within  $4.44e-16$ . The historical Rusanov reference with 257 spatial points differs from this solution by 0.03410 in relative L2 and 0.28578 in maximum error; at 1025 points these discrepancies fall to 0.00733 and 0.06969.

Re-evaluating all 50 frozen V09 Burgers models on the dense grid with Cole–Hopf gives median L2 values 0.30724, 0.26523, 0.22496, 0.26369 and 0.26370 for neither, global\_only, local\_only, full\_exp and full\_quad respectively. No planned contrast passes Holm in this exploratory four-contrast reanalysis. The numerical-reference discrepancy does not explain away the observed large model errors, but it prevents treating historical maximum errors as exact.

## 4.9 Sparse and corrupted observations under a common loss

All 200 runs finish successfully and all 40 five-arm initialization blocks match. The table reports median relative L2 on clean Cole–Hopf targets.

Table 2: Median relative L2 error under difficult observations.

| Case         | Neither | Global only | Local only | Full exp. | Full quad. |
|--------------|---------|-------------|------------|-----------|------------|
| extreme      | 0.53812 | 0.53812     | 0.44694    | 0.45883   | 0.47695    |
| sparse_noisy | 0.46306 | 0.46373     | 0.37333    | 0.37603   | 0.41816    |
| sparse_clean | 0.46061 | 0.45779     | 0.38439    | 0.36866   | 0.40329    |

| Case          | Neither | Global only | Local only | Full exp. | Full quad. |
|---------------|---------|-------------|------------|-----------|------------|
| block_missing | 0.44003 | 0.43793     | 0.34951    | 0.34398   | 0.34288    |

After correction, `full_exp` improves L2 over `global_only` for the missing-block case (paired median difference -0.06999, marginal 95% CI [-0.09386,-0.03466], Holm  $p=0.03125$ ). For extreme data, `full_exp` improves the maximum error over `full_quad` (difference -0.06696, CI [-0.12910,-0.00587], Holm  $p=0.03125$ ). Other planned contrasts do not pass Holm. Section 4.11 separately retests the historical M5/M6/M7/VW comparisons with a common observation loss.

#### 4.10 Doubled optimization budgets

The 200 additional runs use 5000 Burgers, 1200 Allen–Cahn, 3000 Helmholtz and 1200 wave steps. The following median relative L2 values come from independent checkpoint evaluations. Initial tensors match both within each five-arm block and against the historical V09 counterparts. No model selection is based on the evaluation scores.

Table 3: Median relative L2 error at doubled optimization budgets.

| Case       | Neither | Global only | Local only | Full exp. | Full quad. |
|------------|---------|-------------|------------|-----------|------------|
| helmholtz  | 0.08717 | 0.01917     | 0.03245    | 0.00523   | 0.02076    |
| allen_cahn | 0.02859 | 0.02873     | 0.02573    | 0.02785   | 0.02291    |
| burgers    | 0.30754 | 0.24549     | 0.20305    | 0.25680   | 0.26014    |
| wave       | 0.12374 | 0.13976     | 0.02578    | 0.02747   | 0.02652    |

Planned contrasts passing Holm in the doubled-step suite are listed below; the complete table, including null and adverse contrasts, accompanies the code.

Table 4: Planned doubled-budget contrasts passing Holm correction.

| Case      | Metric        | Contrast                 | Paired median difference | Marginal 95% CI         | Holm p  |
|-----------|---------------|--------------------------|--------------------------|-------------------------|---------|
| wave      | relative_l2   | full_exp-<br>global_only | -0.10001                 | [-0.14629,<br>-0.08958] | 0.03125 |
| helmholtz | max_abs_error | interaction              | +0.11349                 | [+0.08393,<br>+0.13949] | 0.03125 |
| wave      | max_abs_error | full_exp-<br>global_only | -0.19593                 | [-0.30415,<br>-0.14881] | 0.03125 |

The quadratic-potential L2 advantage established for the V09 wave configuration does not pass Holm at the doubled horizon. `Full_exp` still improves both L2 and maximum error over `global_only` for waves. A positive Helmholtz maximum-error interaction reflects the additive contrast and does not imply that `full_exp` is the least accurate model.

The doubled-step comparison measures sensitivity to optimization duration. It is not an equal-compute or equal-time experiment, and historical and V10 wall times must not be pooled as a controlled timing comparison.

#### 4.11 Re-running the historical baselines with a common observation loss

An additional 160 runs cover M5, M6, M7 and VW across the same four observation regimes and ten seeds, using the exact V10 MSE/Cole–Hopf configurations. Forty full\_exp checkpoints from Section 4.9 are reused as vRBA comparators; they are not new or independent training runs. Initial parameter and observation tensors match across all five compared methods. The table gives median L2.

Table 5: Common-loss baseline median relative L2 errors.

| Case          | M5      | M6      | M7      | VW      | vRBA    |
|---------------|---------|---------|---------|---------|---------|
| block_missing | 0.56166 | 0.55393 | 0.53397 | 0.48595 | 0.34398 |
| sparse_clean  | 0.58673 | 0.55823 | 0.55471 | 0.49238 | 0.36866 |
| sparse_noisy  | 0.57248 | 0.55808 | 0.54189 | 0.50375 | 0.37603 |
| extreme       | 0.61185 | 0.60330 | 0.59146 | 0.56037 | 0.45883 |

All newly trained checkpoints have been independently re-evaluated. The following contrasts pass Holm in the separately declared family of 16 per metric. All contrasts, including null and unfavorable effects, are retained in the accompanying paired\_statistics.csv.

Table 6: Common-loss paired contrasts passing Holm correction.

| Case          | Metric      | Contrast | Paired median difference | Marginal 95% CI      | Holm p  |
|---------------|-------------|----------|--------------------------|----------------------|---------|
| block_missing | relative_l2 | vrba-m5  | -0.19580                 | [-0.24059, -0.18515] | 0.03125 |
| block_missing | relative_l2 | vrba-m6  | -0.19318                 | [-0.24042, -0.18069] | 0.03125 |
| block_missing | relative_l2 | vrba-m7  | -0.17172                 | [-0.20840, -0.12736] | 0.03125 |
| block_missing | relative_l2 | vrba-vw  | -0.13158                 | [-0.15465, -0.09959] | 0.03125 |
| sparse_clean  | relative_l2 | vrba-m5  | -0.22439                 | [-0.32713, -0.15273] | 0.03125 |
| sparse_clean  | relative_l2 | vrba-m6  | -0.21344                 | [-0.32406, -0.15757] | 0.03125 |
| sparse_clean  | relative_l2 | vrba-m7  | -0.17495                 | [-0.24497, -0.13279] | 0.03125 |
| sparse_noisy  | relative_l2 | vrba-m5  | -0.19789                 | [-0.30269, -0.17423] | 0.03125 |
| extreme       | relative_l2 | vrba-m5  | -0.13299                 | [-0.20818, -0.06938] | 0.03125 |
| extreme       | relative_l2 | vrba-m6  | -0.13806                 | [-0.20866, -0.11213] | 0.03125 |

| Case    | Metric      | Contrast | Paired median difference | Marginal 95% CI      | Holm p  |
|---------|-------------|----------|--------------------------|----------------------|---------|
| extreme | relative_l2 | vrba-m7  | -0.13349                 | [-0.15551, -0.06411] | 0.03125 |

vrBA has the lowest median L2 in all four regimes, with 11 of 16 L2 contrasts passing Holm; no maximum-error contrast passes its separate correction. This targeted rerun removes the observation-loss mismatch in these cases. Its protocol followed the V10 audit and remains exploratory. It does not establish performance on unseen physical systems or a general robustness guarantee.

The following figures are generated from the stored numerical analyses. Figure 1 documents the Burgers reference discrepancy; Figure 2 displays every paired seed in the common-loss comparison; Figure 3 shows doubled-step ablations. They must be interpreted together with the corrected statistics.

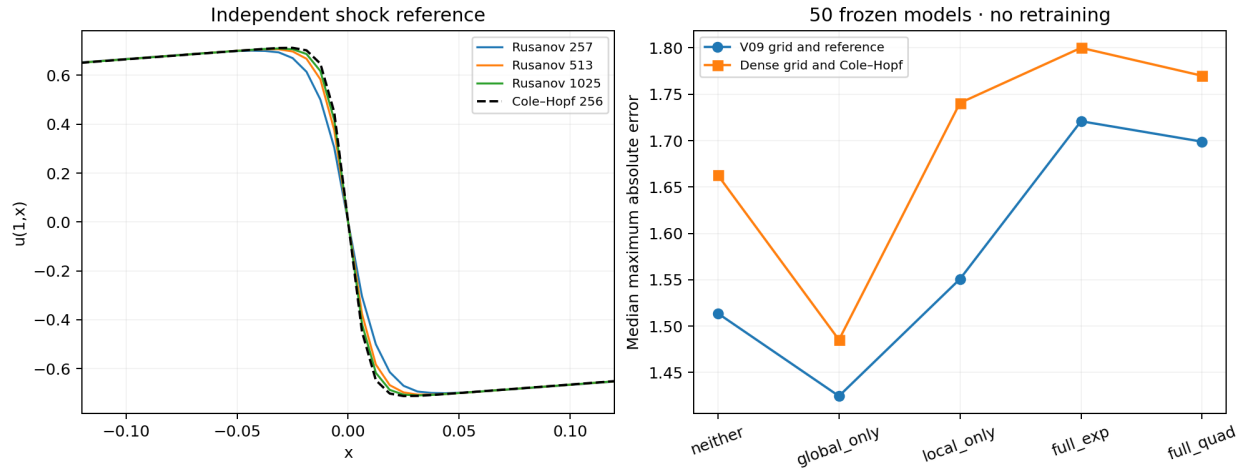


Figure 1: Burgers reference validation and re-evaluation of frozen V09 models. The second panel also changes the evaluation grid; it does not isolate the reference change alone.

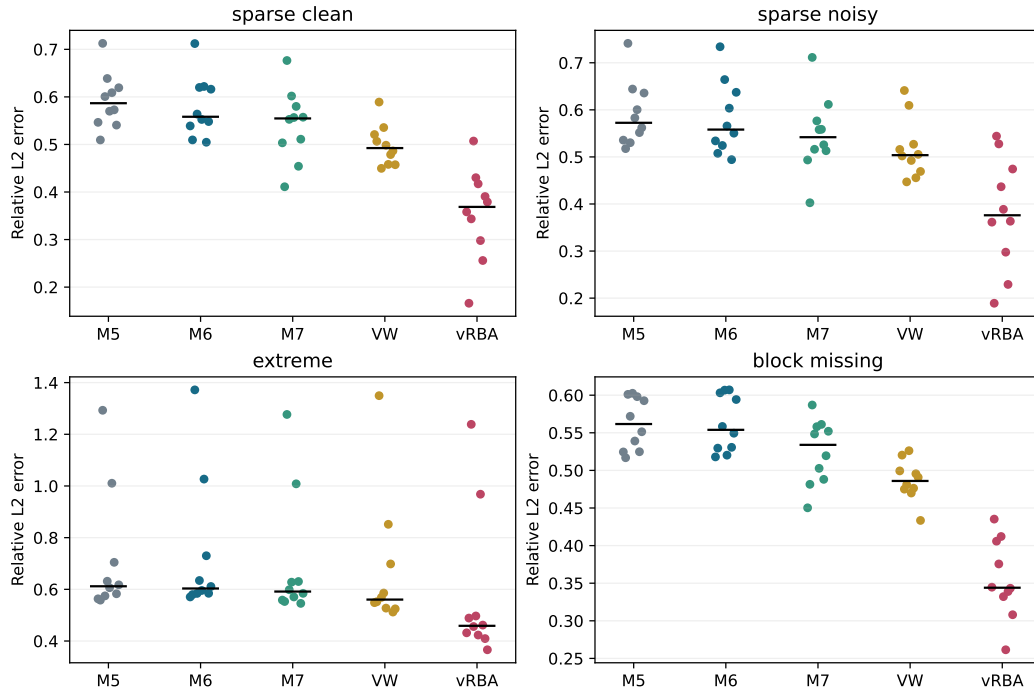


Figure 2: Common-MSE Burgers comparisons. Points represent the ten individual seeds and black bars show medians. No inference is based solely on these visual rankings.

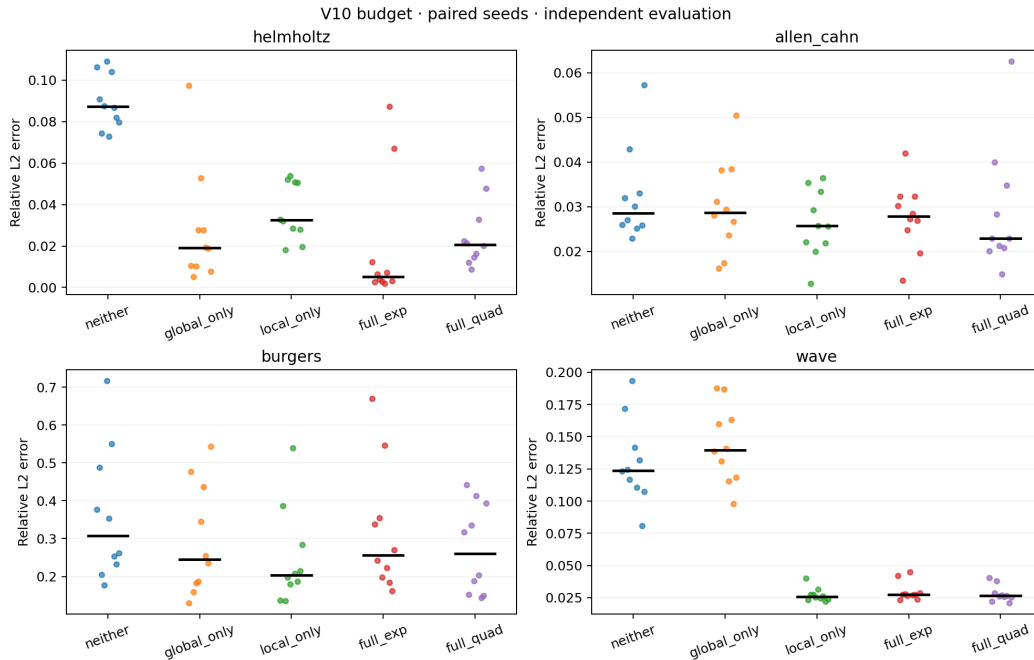


Figure 3: Doubled-step ablations across four PDE configurations. Budgets match optimization steps within each PDE and do not equalize wall time.

## 5 Discussion

The experiments evaluate the original fixed-reference hypothesis and separate three statements that are often conflated. First, adaptive collocation can change what a loss controller observes. Second, removing this change can alter gradients. Third, neither fact guarantees lower solution error. M6 demonstrates that an independent reference set can change the controller without producing a robust benefit. M7 demonstrates that exact proposal correction can restore a target finite-population mean and improve a matched control in data-poor Burgers regimes. VW demonstrates that an estimated geometric correction can be equally or more effective, especially when its bandwidth is favorable and global component weights remain fixed.

The poor behavior of VW-CA relative to VW in several cases suggests an interaction between local volume weights and global component balancing. One possible explanation is that squared local volume weights change the magnitude and variance of the physics gradient, after which the global controller reacts to a signal that is already strongly rescaled. This interpretation is an inference, not yet a demonstrated mechanism. A factorial experiment varying local correction, global controller and update frequency is needed.

The vRBA comparison sharpens the distinction between restoring a target measure and deliberately changing the training objective. In this benchmark, the tilted objective is much more effective in global L2 error than either exact or KDE-based correction. Yet the Burgers maximum-error result shows that this benefit is not uniform over the domain. The spatial maps document localized error patterns alongside lower global error. They do not establish that attention causes the localized spike. The V09–V10 ablations vary local/global mechanisms under fixed point sets, while the historical recipe comparisons also differ in sampling and controller details.

The practical conclusion is not that one correction should replace every other. When the proposal probability is exactly available, M7 avoids density estimation and has favorable asymptotic correction cost. When only a realized nonuniform point cloud is available, volume weighting remains applicable. If the objective is deliberately tilted toward high residuals rather than restored to a uniform measure, a variational adaptive method such as vRBA may be more appropriate.

## 6 Limitations

1. The VW implementation reproduces its core weighting mechanism on a common backbone, not the complete published architecture and Adam–L-BFGS schedule.
2. vRBA is implemented on the common backbone, but the complete official JAX architecture, hard constraints and long optimizer schedule have not been reproduced.
3. The original high-frequency Helmholtz case remains unsolved; the calibrated case answers a different, easier question and is reported separately.
4. Ten seeds provide useful paired evidence but limited power for small effects.
5. KDE bandwidth sensitivity was tested on two problems, not the full suite.
6. Timing conclusions are CPU-specific, and the KDE scaling claim still needs controlled multi-size and GPU measurements.
7. The PDE suite is synthetic. No conclusion currently transfers to tokamak plasma data or real-time instability control.

8. Local/global and potential ablations, spatial maps and doubled-step sensitivity are available. Common-loss noisy baseline reruns are also available. Equal-time comparisons remain outside the claims of this paper. A separate-host portability protocol is supplied; its execution status is reported in the supplement.
9. The public literature review is reproducible but not exhaustive across subscription databases.

## 7 Conclusion

Residual sampling and adaptive loss balancing form a feedback system whose behavior depends on the measure used to estimate each loss component. The tested fixed-reference ReCoA-PINN implementation does not establish general stabilization across the evaluated configurations. Exact proposal reweighting can reduce distribution-induced gradient discrepancies and improve matched controls in sparse/noisy Burgers regimes, but it does not dominate KDE volume weighting across PDEs or bandwidths. The common-backbone vRBA recipe shows strong historical relative-L2 performance, qualified by a noisy-data loss confound and substantial maximum-error and cost tradeoffs. The new common-loss ablations and doubled budgets identify conditional rather than universal benefits. Robust claims about adaptive PINNs therefore require matched controls, several PDEs and seeds, multiple error metrics, sensitivity analyses, and explicit separation between restoring a measure and optimizing a deliberately tilted objective.

## 8 Data, code and reproducibility statement

The PyTorch implementation stores resolved YAML configurations, software and hardware metadata, metric trajectories, model checkpoints and final summaries for every run. Campaign manifests regenerate paired tables and figures while resuming only missing cells. The repository contains automated unit and smoke tests for samplers, corrections, PDE residuals, reproducibility and reporting. V09 supplies 200 ablation runs and V10 adds 560 verified runs, in addition to the historical baseline campaigns. The V10 numerical suite passed 55 tests. Raw campaign counts distinguish completed training, repeated profiling and reused historical evidence; they must not be summed as independent physical experiments.

## 9 Independent-host verification

On 6 September 2026, a separate GitHub-hosted Ubuntu runner reconstructed the scores of all 560 frozen V10 models and all 96 paired contrasts within the predeclared tolerance (relative  $10^{-8}$ , absolute  $10^{-10}$ ). The 55 automated tests passed. All 12 preselected retrainings completed and their L2 and maximum errors met the predeclared numerical tolerance (relative  $10^{-3}$ , absolute  $10^{-5}$ ). The largest absolute differences were  $9.53 \times 10^{-9}$  in L2 and  $1.40 \times 10^{-8}$  in maximum error.

The stricter joint gate nevertheless failed: initial model fingerprints matched in all 12 cases, but training/reference-set fingerprints differed in the four noisy-observation cases. Only 8/12 satisfied every declared condition. The environments differ in host and PyTorch version (2.14.0+cpu versus 2.8.0+cpu). The original raw initial tensors were not retained, so the differing bytes cannot be located retrospectively. A floating-point or library-version effect is plausible but not established by these hashes. This is successful numerical reconstruction with a documented input-identity limitation, not

bitwise reproduction of the complete experiment. The failed gate and its thresholds are preserved in the repository and supplement.

## 10 Author metadata and declarations

This is an author-review copy. Youssef EL MOUTEE is the identified working author. The final author list and order, affiliations, corresponding-author details, contributions, funding and competing-interest statements require author confirmation before submission. No unconfirmed affiliation or consent has been assigned to the supervisor. The code and archived evidence are in the public repository [ucef90/Doctorat](#), with frozen V10 evidence at commit `4161ed57f232ea7b796480b38201502b3cabbf67`. All journal-specific declarations require author confirmation before submission.

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